

Debian at Yeeloong (MIPS)

You can find a number of alternative solutions, which, like ARM, can be quite energy efficient, and suitable for the mobile sector. One bright example is the Yeeloong 8089B netbook from Lemote. It has 1 GB of installed RAM too, a 6.35 cm (2.5") HDD, but only of 160 GB. It features a Loongson 2F processor at 800MHz. It was manufactured a few years earlier than the Nvidia Tegra2, so please don't expect any hardware acceleration because that has only appeared in the Loongson 2H (aka Godson-2H). So, our testing is without hardware acceleration in mind. I can't wait to get my hands on the next-generation Yeeloong 8133, designed on the Loongson 3A.

```
loongson@debian:~$ time tz b
```

```
7-Zip 9.04 beta Copyright (c) 1999-2009 Igor Pavlov 2009-06-30
p7zip Version 9.04 (locale=C,Utf16=off,HugeFiles=on,1 CPU)
```

```
RAM size: 1004 MB, # CPU hardware threads: 1
RAM usage: 419 MB, # Benchmark threads: 1
```

Dict	Compressing				Decompressing			
	Speed Usage	R/U Rating			Speed Usage	R/U Rating		
	KB/s	%	MIPS	MIPS	KB/s	%	MIPS	MIPS
22:	315	99	309	306	4989	99	454	450
23:	312	99	321	318	4937	99	455	452
24:	303	99	329	326	4880	99	457	452
25:	297	99	342	339	4839	99	459	455

Avr:	99	325	322		99	456	452	
Tot:	99	391	387					

```

real 4m38.692s
user 4m31.901s
sys 0m3.776s
```

Compared with the TrimSlice, this is slower. But remember, it is not contemporary. Also, the Nvidia Tegra2 has two virtual cores, while the Loongson2F has only one.

Next is the hard disk partition. It has one large ext3 root partition of 60 GB. The remaining space is reserved, but can be manually added and formatted as new partitions.

```
loongson@debian:~$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/hda1	56G	53G	202M	100%	/
tmpfs	583M	0	583M	0%	/lib/init/rw
udev	10M	432K	9.6M	5%	/dev
tmpfs	583M	0	583M	0%	/dev/shm

```

loongson@debian:~$ mount
/dev/hda1 on / type ext3 (rw,errors=remount-ro)
```



Figure 4: SunSpider benchmark, Firefox 9



Figure 5: Chromium with SunSpider



Figure 6: Flash-player works on Linux/ARM!



Figure 7: MIPS-based netbook from Lemote

```

tmpfs on /lib/init/rw type tmpfs (rw,nosuid,nodev)
proc on /proc type proc (rw,noexec,nosuid,nodev)
sysfs on /sys type sysfs (rw,noexec,nosuid,nodev)
udev on /dev type tmpfs (rw,nodev)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev)
devpts on /dev/pts type devpts (rw,noexec,nosuid,gid=5,mode=620)
fusectl on /sys/fs/fuse/connections type fusectl (rw)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc
(rw,noexec,nosuid,nodev)
```

This unit shipped with Debian Lenny by default. Three years have passed, so I decided to update it to Debian Squeeze. The operation requires two steps: a package list update and upgrade option, after editing repo information to add new 'Squeeze' lines into `/etc/apt/sources.list`, and comment out old ones:

```

#deb http://ftp.uk.debian.org/debian/ lenny main contrib non-free
deb http://ftp.uk.debian.org/debian/ squeeze main contrib non-free
```

Now, let's run `aptitude update` && `aptitude safe-upgrade`. Once the update is done, it's worth doing a reboot.

Web browsers for MIPS

The situation with MIPS is slightly worse than with ARM. First, there is no entity like Linaro/Canonical that would push software development for the MIPS Linux port. Yes, Lemote has a nice manufacturing basis. However, in terms of the optimal software development stack's speed, the number of